⬡ POWER BI · INDIA EDITION

DAX

THE COMPLETE LEARNING GUIDE

for Power BI Data Analysts

Dataset ▶ **Indian E-commerce · 3 Tables · Real Data**

30+ Functions | 10 Real KPIs | 30 Interview Q&As

30-Day Roadmap | 9 Sections | Full Project Build

From Excel to Power BI in 30 Days.

Built for Indian freshers. No coding background needed. Every formula tied to a real business scenario your manager will actually ask for.

★★★★★ Interview-Ready · Classroom-Proven



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SECTION 1: INTRODUCTION TO DAX

What is DAX — And Why Should You Care?

Let me start with a question. Imagine you are a Data Analyst at Flipkart. Your manager walks in on a Monday morning and says: 'I need to know which state gave us the highest profit last quarter, and how does that compare to the same quarter last year?'

Can you answer that using just Excel formulas? Maybe. But what if you have 10 lakh rows of order data? Excel will crash. What if the data refreshes every day automatically? Excel won't handle that either. This is exactly why DAX was created.

Full Form and Origin

DAX stands for Data Analysis Expressions. Microsoft introduced DAX in 2010 with PowerPivot, and later built it into the heart of Power BI. Think of DAX as the language that Power BI uses to calculate, summarize, and analyse data intelligently — the way SQL is the language of databases.

How DAX is Different from Excel Formulas

Feature	Excel Formula	DAX Measure
Calculation Scope	Fixed cell or range	Entire column / table
Filter Response	Manual — you change the range	Automatic — responds to report filters
Data Volume	1 million row limit	Handles 100M+ rows efficiently
Date Calculations	Basic functions	Full Time Intelligence library
Cross-Table Calc	Complex VLOOKUP chains	Clean RELATED / CALCULATE syntax
Performance	Slow on large data	In-memory engine, very fast

Why Companies Use DAX

- Automate reporting that used to take hours in Excel
- Create self-service dashboards for non-technical managers
- Perform complex time-based comparisons (YoY, MoM, YTD)
- Build KPI scorecards that update automatically with new data
- Combine data from multiple sources into unified insights

DAX Across Industries — Real Examples

Industry	Company Examples	DAX Use Case
E-commerce	Flipkart, Meesho, Myntra	Daily sales, return rates, top sellers, customer LTV
Retail	DMart, Reliance Retail	Store performance, inventory turnover, promotion ROI
Banking	HDFC, ICICI, SBI	Loan disbursement trends, NPA ratios, branch ranking

Industry	Company Examples	DAX Use Case
Healthcare	Apollo, Fortis, Narayana	Patient volumes, revenue per department, doctor KPIs
EdTech	BYJU'S, Unacademy	Course completion, revenue per student, batch analysis

Business Questions That CANNOT Be Answered Without DAX

1. What was our revenue growth compared to the same month last year?
2. Which customers have ordered more than 3 times?
3. What percentage of total sales does each category contribute?
4. What is the profit margin after removing returned orders?
5. Which product was the top seller in each state separately?
6. What is the 3-month rolling average of daily sales?

ℹ BUSINESS INSIGHT

All 6 questions above are asked in real business review meetings every month. A Data Analyst who can answer them independently — without waiting for IT to run SQL — is genuinely valuable.

SECTION 2: MEASURES vs CALCULATED COLUMNS vs CALCULATED TABLES

This is the section where most beginners get confused. The difference is subtle but critical. Get this right and everything else becomes easier.

What is a Measure?

A Measure is a dynamic calculation. It calculates based on whatever filters are applied in the report at that moment. You write it once, and Power BI recalculates it every time the filter context changes.

Think of a Measure like a smart formula that constantly asks: 'What is the total right now, given the current filters?'

```
Total Sales = SUM(Orders[Sales Amount])

-- If State filter = Maharashtra  →  shows Maharashtra sales
-- If Category filter = Electronics  →  shows Electronics sales
-- Same formula, different results based on context
```

What is a Calculated Column?

A Calculated Column is a static, row-level calculation. It calculates once when you create it and stores the result as a new column in your table, row by row. Like adding a formula column in Excel.

```
Profit = Orders[Sales Amount] - Orders[Cost Amount]

-- Row 1: 2500 - 1800 = 700  (stored)
-- Row 2: 5000 - 3200 = 1800  (stored)
-- Does NOT change when you apply report filters
```

What is a Calculated Table?

A Calculated Table is a completely new table created using DAX. Used for date tables, summary tables, or helper tables for specific analyses.

```
CategorySummary = SUMMARIZE(
    Orders,
    Products[Category],
    "Total Sales", SUM(Orders[Sales Amount])
)
```

Decision Rule — Which One to Use?

Scenario	Use This
Number on a KPI card that changes with slicers	Measure
Percentage shown in a chart that filters by region	Measure

Scenario	Use This
New column categorizing each order as High/Medium/Low	Calculated Column
Adding product category to each order row	Calculated Column
Creating a Date table for time intelligence	Calculated Table
Summary table for a disconnected slicer	Calculated Table

Scenario Walkthroughs — Wrong vs Correct Approach

Scenario 1: Show Total Sales on a Dashboard

Business Requirement: Sales total that changes when user selects a state.

> ⚠ **COMMON MISTAKE**
> WRONG: Create a Calculated Column 'Sales_Col = Orders[Sales Amount]'. This just copies the column. It shows individual row values, not a total. And it does NOT respond to filters.

> **PRO TIP**
> CORRECT: Create a Measure 'Total Sales = SUM(Orders[Sales Amount])'. Measures aggregate data and respond to filters — exactly what dashboards need.

Scenario 2: Categorize Each Order by Value

Business Requirement: Label each order as High/Medium/Low value for segmentation.

> ⚠ **COMMON MISTAKE**
> WRONG: Try to do this with a Measure. Measures produce one aggregated value, not a row-level label for each order.

> **PRO TIP**
> CORRECT: Create a Calculated Column in the Orders table.

```
Order Category = IF(
    Orders[Sales Amount] > 5000, "High Value",
    IF(Orders[Sales Amount] >= 2000, "Medium Value", "Low Value")
)
```

SECTION 3: DAX FUNDAMENTALS

Before writing serious DAX formulas, you need to understand how DAX thinks. This is the grammar of DAX.

Syntax Structure

```
Measure Name = FUNCTION(Arguments)

-- Example:
Total Sales = SUM(Orders[Sales Amount])

-- Breaking it down:
-- 'Total Sales'          → Name you give this measure
-- '='                    → Assignment operator
-- 'SUM'                  → DAX function
-- 'Orders[Sales Amount]' → Argument (TableName[ColumnName])
```

Column Reference Format

You always reference columns as TableName[ColumnName]. This is different from Excel where you use column letters like C2:C100.

Reference	Example
Orders[Sales Amount]	Sales Amount column from Orders table
Customers[Customer Name]	Customer Name from Customers table
Products[Category]	Category from Products table

DAX Data Types

Data Type	Examples	Common Columns
Whole Number	5, 100, 2024	Quantity, Order Count, Year
Decimal Number	2500.50, 34.67	Sales Amount, Profit Margin
Text	'Maharashtra', 'Electronics'	State, Category, Customer Name
Date/Time	01-Jan-2024	Order Date, Ship Date
Boolean	TRUE, FALSE	Conditions in IF statements
Blank	(empty)	Missing values, unmatched rows

How DAX Evaluates Formulas

DAX does not evaluate formulas the way Excel does. In Excel, you press Enter and the formula runs on a specific cell. In DAX, the formula runs in context — the context is determined by filters active on your report.

When you have a Card visual showing Total Sales with a State slicer set to 'Maharashtra', DAX evaluates your formula with Maharashtra as the filter context. Change the slicer to 'Karnataka' and DAX re-evaluates automatically.

> **BUSINESS INSIGHT**
> This automatic context awareness is what makes DAX powerful. One formula adapts to thousands of filter combinations without any manual adjustment.

Handling Blank Values

DAX treats BLANK differently from zero. SUM of blank values returns BLANK, not zero. This matters in division — dividing by BLANK can cause errors. Always use the DIVIDE function instead of the / operator.

```
-- UNSAFE: Can error when denominator is blank or zero
Margin = [Total Sales] / [Total Cost]

-- SAFE: Returns 0 instead of error
Margin = DIVIDE([Total Sales], [Total Cost], 0)
```

SECTION 4: MOST IMPORTANT DAX FUNCTIONS FOR FRESHERS

This is the main skill-building section. Every function is demonstrated using the ShopIndia e-commerce dataset. Master these and you can handle 90% of real business DAX requirements.

1. SUM — Total Revenue

The most fundamental calculation in any business dashboard.

Business Use Case

The business head wants total sales revenue on the executive dashboard. This is always the first KPI.

```
Total Sales = SUM(Orders[Sales Amount])

-- Result: ₹45,67,823 (grand total, no filter)
-- With Maharashtra filter: ₹12,34,500
-- With Electronics filter: ₹18,90,200
```

ℹ BUSINESS INSIGHT

If Total Sales = ₹45.6L, management compares against ₹50L target. Gap of ₹4.4L triggers regional review.

⚠ COMMON MISTAKE

Using SUM on a text column gives an error. Using SUM in a Calculated Column instead of a Measure means it won't respond to filters.

INTERVIEW QUESTION

Q: What is the difference between SUM and SUMX?

A: SUM adds an existing column directly. SUMX iterates row by row, evaluates an expression per row, then sums. Use SUMX when you need to calculate something per row first — e.g., SUMX(Orders, Orders[Sales Amount] * 1.18) for GST-inclusive total.

2. AVERAGE — Typical Transaction Size

Total sales tells the big picture. Average tells you the typical deal size.

```
Average Order Value = AVERAGE(Orders[Sales Amount])

-- Or using measures (preferred):
Average Order Value = DIVIDE([Total Sales], [Total Orders], 0)
```

ℹ BUSINESS INSIGHT

If Average Order Value = ₹1,240, the logistics team sets free delivery threshold at ₹999 to encourage larger baskets.

⚠ COMMON MISTAKE

AVERAGE includes zero-value rows. Use AVERAGEX with FILTER to exclude zeros if needed.

INTERVIEW QUESTION

Q: How do you calculate average sales only for orders above ₹500?

A: Average High Value = AVERAGEX(FILTER(Orders, Orders[Sales Amount] > 500), Orders[Sales Amount])

3. MIN and MAX — Range and Extremes

```
Earliest Order Date = MIN(Orders[Order Date])
Latest Order Date   = MAX(Orders[Order Date])
Highest Single Order = MAX(Orders[Sales Amount])
Lowest Single Order  = MIN(Orders[Sales Amount])
```

ℹ BUSINESS INSIGHT

Highest Single Order = ₹89,500 may flag a VIP bulk order worth personal follow-up from the sales team.

INTERVIEW QUESTION

Q: What is the difference between MAX() and MAXX()?

A: MAX(column) returns the maximum value in a column. MAXX(table, expression) evaluates expression row by row and returns the maximum result. Example: MAXX(Orders, Orders[Sales Amount] * 1.18) gives the maximum GST-inclusive sale.

4. COUNT and DISTINCTCOUNT — Order vs Customer Count

```
Total Orders    = DISTINCTCOUNT(Orders[Order ID])
Total Customers = DISTINCTCOUNT(Orders[Customer ID])
Total Products Sold = DISTINCTCOUNT(Orders[Product ID])

-- COUNT counts rows with numeric values
-- DISTINCTCOUNT counts UNIQUE values only
```

Metric	Function	Result	Business Meaning
Order Lines	COUNT(Orders[Order ID])	1,450	Total order line items

Metric	Function	Result	Business Meaning
Unique Orders	DISTINCTCOUNT(Orders[Order ID])	1,200	Actual transactions placed
Unique Customers	DISTINCTCOUNT(Orders[Customer ID])	380	Active customer base

ℹ BUSINESS INSIGHT

1,200 orders from only 380 customers = average 3.2 orders per customer. Strong retention. Focus on acquisition.

INTERVIEW QUESTION

Q: A customer placed 5 orders. How do COUNT and DISTINCTCOUNT treat this?

A: COUNT adds 5 to the total (one per order row). DISTINCTCOUNT adds 1 to the unique customer count.

5. IF and SWITCH — Conditional Logic

Real business data needs categorization. IF and SWITCH create labels, tiers, and conditional outputs.

IF — Two Outcomes

```
-- Calculated Column: Profit Status
Profit Status = IF(
    Orders[Sales Amount] > Orders[Cost Amount],
    "Profitable",
    "Loss"
)
```

SWITCH — Multiple Conditions (Cleaner than nested IFs)

```
-- Calculated Column: Region
Region = SWITCH(Orders[State],
    "Maharashtra", "West",
    "Gujarat",     "West",
    "Karnataka",   "South",
    "Tamil Nadu",  "South",
    "Delhi",       "North",
    "Uttar Pradesh", "North",
    "Other"       -- default
)
```

ℹ BUSINESS INSIGHT

Region column enables a regional breakdown bar chart. West vs South vs North vs East — one column powers the entire regional analysis.

⚠ COMMON MISTAKE

Nesting more than 3 IFs becomes unreadable. Use SWITCH for multiple conditions on the same column.

INTERVIEW QUESTION

Q: When use SWITCH instead of IF?

A: When you have 3+ conditions based on the same column's exact values. SWITCH is cleaner and faster to read. IF is better for range-based conditions like greater than or less than.

6. LEFT, RIGHT, MID, LEN — Text Extraction

```
-- Order ID format: 'ORD-2024-00123'

-- Extract year portion
Order Year Part = MID(Orders[Order ID], 5, 4)    -- returns '2024'

-- Extract order number
Order Number = RIGHT(Orders[Order ID], 5)         -- returns '00123'

-- Extract prefix
ID Prefix = LEFT(Orders[Order ID], 3)             -- returns 'ORD'

-- Check city name length (data quality)
City Name Length = LEN(Orders[City])
```

⚠ COMMON MISTAKE

LEN counts spaces too. 'Mumbai ' with trailing space has LEN = 7, not 6. MID positions start at 1, not 0.

INTERVIEW QUESTION

Q: How do you extract state code from 'MH-Mumbai'?

A: LEFT("MH-Mumbai", 2) returns 'MH'. For dynamic prefix extraction before hyphen, use LEFT(text, FIND("-", text) - 1).

7. Date Functions — TODAY, YEAR, MONTH, QUARTER, DATEDIFF

```
-- Add time dimensions to Orders table (Calculated Columns)
Order Year    = YEAR(Orders[Order Date])
```

```
Order Month   = MONTH(Orders[Order Date])
Month Name    = FORMAT(Orders[Order Date], "MMMM")
Order Quarter = QUARTER(Orders[Order Date])

-- Measure: Days since last order
Days Since Last Order = DATEDIFF(MAX(Orders[Order Date]), TODAY(), DAY)

-- Calculated Column: Order processing time
Processing Days = DATEDIFF(Orders[Order Date], Orders[Ship Date], DAY)
```

ℹ BUSINESS INSIGHT

Processing Days average of 4.2 days with a spike to 9 days in December signals a holiday season bottleneck for operations to investigate.

⚠ COMMON MISTAKE

DATEDIFF(start, end, interval) — if start and end are reversed, you get negative values. MONTH() returns 1–12 (number). Use FORMAT() for month names.

INTERVIEW QUESTION

Q: How do you calculate a customer's age from date of birth?

A: Customer Age = DATEDIFF(Customers[Date of Birth], TODAY(), YEAR)

8. DIVIDE and ROUND — Safe Math

```
-- UNSAFE approach (errors when cost = 0)
Margin = [Total Sales] / [Total Cost]

-- SAFE approach with DIVIDE
Profit Margin % = DIVIDE(
    [Total Sales] - [Total Cost],
    [Total Sales],
    0        -- alternate result when denominator = 0
) * 100

-- ROUND to 2 decimal places
Margin Rounded = ROUND([Profit Margin %], 2)
```

⚠ COMMON MISTAKE

Always use DIVIDE instead of / in measures. One period with zero sales will break your entire visual if you use /.

INTERVIEW QUESTION

Q: What happens if you use / and Cost Amount is zero for some period?

A: It returns an error or BLANK, which breaks your visual. DIVIDE() returns the alternate result (usually 0 or BLANK by your choice), keeping the visual intact.

9. FILTER, ALL, VALUES — Context Control

```
-- FILTER: Count only qualifying rows
High Value Orders = COUNTROWS(FILTER(Orders, Orders[Sales Amount] > 5000))

-- ALL: Remove filter to get grand total (used in % of total)
Category % of Total = DIVIDE(
    [Total Sales],
    CALCULATE([Total Sales], ALL(Products[Category])),
    0
) * 100

-- VALUES: Unique values respecting current filter
Selected States = CONCATENATEX(VALUES(Orders[State]), Orders[State], ", ")
```

ℹ BUSINESS INSIGHT

Category % of Total shows Electronics = 38%, Fashion = 29%, Home = 21%, Others = 12%. Management allocates next marketing budget proportionally.

INTERVIEW QUESTION

Q: What is the difference between ALL and ALLEXCEPT?

A: ALL removes all filters from the specified table/column. ALLEXCEPT removes all filters EXCEPT the ones you specify — useful for 'share within region' calculations.

10. RELATED and RELATEDTABLE — Cross-Table Data

```
-- RELATED: Bring value from parent table into child table
-- Calculated Column in Orders table:
Product Category = RELATED(Products[Category])
Customer Segment = RELATED(Customers[Segment])

-- RELATEDTABLE: Count rows from child table in parent table
-- Calculated Column in Customers table:
Order Count = COUNTROWS(RELATEDTABLE(Orders))
Total Spent = SUMX(RELATEDTABLE(Orders), Orders[Sales Amount])
```

Function	Direction	Use When
RELATED	Many → One (child → parent)	Bring a single attribute from parent
RELATEDTABLE	One → Many (parent → child)	Aggregate child records in parent

⚠ COMMON MISTAKE

RELATED only works from the many-side to the one-side. If relationships are not set up in Model view, RELATED will error.

INTERVIEW QUESTION

Q: Can you use RELATED in a Measure?

A: RELATED is designed for Calculated Columns (row context). In Measures, use LOOKUPVALUE or CALCULATE with FILTER for cross-table lookups.

11. RANKX — Rankings and Top N

```
-- Rank states by total sales (Rank 1 = highest)
State Sales Rank = RANKX(
    ALL(Orders[State]),           -- rank across all states
    CALCULATE(SUM(Orders[Sales Amount])),  -- rank by this value
    ,                             -- (value argument, leave blank)
    DESC,                         -- highest gets rank 1
    Dense                         -- no gaps (1,2,3,3,4 not 1,2,3,3,5)
)
```

ℹ BUSINESS INSIGHT

Add State Sales Rank to a table visual. Filter where Rank <= 5 to create an auto-updating Top 5 States dashboard element.

INTERVIEW QUESTION

Q: What is the difference between Dense and Skip ranking?

A: Dense: 1,2,3,3,4 (no gaps after ties). Skip: 1,2,3,3,5 (skips rank 4 when two items tie at 3). Dense is standard for business dashboards.

12. SUMX and AVERAGEX — Iterator Functions

```
-- Total Revenue including 18% GST
Total Revenue with GST = SUMX(Orders, Orders[Sales Amount] * 1.18)

-- How SUMX works row by row:
```

```
-- Row 1: 1000 * 1.18 = 1180
-- Row 2: 2500 * 1.18 = 2950
-- Row 3: 500  * 1.18 = 590
-- SUMX total: 4720

-- Average profit per order
Average Profit Per Order = AVERAGEX(
    Orders,
    Orders[Sales Amount] - Orders[Cost Amount]
)
```

ℹ BUSINESS INSIGHT

SUMX and AVERAGEX are essential when you need to calculate an expression at the row level before aggregating. You cannot do this with SUM or AVERAGE alone.

⚠ COMMON MISTAKE

If you write SUMX(Orders, Orders[Sales Amount]) with no row-level calculation, it is identical to SUM(Orders[Sales Amount]). SUMX has overhead — only use it when needed.

INTERVIEW QUESTION

Q: When use SUMX instead of SUM?

A: When you need to calculate something per row before summing. SUM works on an existing column. SUMX works on an expression computed per row — like quantity × unit price, or sales × tax rate.

13. Time Intelligence — TOTALYTD, SAMEPERIODLASTYEAR, DATEADD

Time intelligence is what separates a basic Power BI report from a real business intelligence dashboard. Requires a proper Date table.

```
-- Year-to-Date cumulative sales
YTD Sales = TOTALYTD(SUM(Orders[Sales Amount]), Orders[Order Date])

-- Same period last year
Last Year Sales = CALCULATE(
    [Total Sales],
    SAMEPERIODLASTYEAR(Orders[Order Date])
)

-- Year-over-Year growth %
YoY Growth % = DIVIDE(
    [Total Sales] - [Last Year Sales],
    [Last Year Sales],
    0
) * 100

-- Sales 3 months ago
```

```
Sales 3 Months Ago = CALCULATE(
    [Total Sales],
    DATEADD(Orders[Order Date], -3, MONTH)
)
```

Function	What It Does	Business Use Case
TOTALYTD	Cumulative total from Jan 1 to current date	Track progress toward annual target
SAMEPERIODLASTYEAR	Same date range in the previous year	YoY comparison in every visual
DATEADD	Shift date range by N periods	Compare current vs 3 months ago

⚠ COMMON MISTAKE

Time intelligence requires a Date table marked as 'Date Table' in Power BI with a relationship to Orders[Order Date]. Without this, results are incorrect or BLANK.

INTERVIEW QUESTION

Q: Why do time intelligence functions require a Date table?

A: These functions need a continuous, gap-free calendar to compute period boundaries correctly. If your sales dates have gaps (no orders on weekends), TOTALYTD gets confused without a complete calendar.

SECTION 5: BUILDING REAL BUSINESS KPIs

Now that you know individual functions, let us build the KPIs that appear on a real business dashboard. These are measures your manager will ask for on Day 1.

KPI	DAX Formula	Visual	Management Decision
Total Sales	SUM(Orders[Sales Amount])	KPI Card	Compare against target; find shortfall
Total Orders	DISTINCTCOUNT(Orders[Order ID])	KPI Card	Volume trend; B2B vs B2C signal
Total Customers	DISTINCTCOUNT(Orders[Customer ID])	KPI Card	Acquisition vs retention balance
Total Profit	[Total Sales] - [Total Cost]	KPI Card (conditional format)	Rising costs flag with stable sales
Profit Margin %	DIVIDE([Profit], [Total Sales], 0)*100	Gauge vs target	Drop below 20% triggers pricing review
Avg Order Value	DIVIDE([Total Sales], [Total Orders], 0)	KPI with trend	Low AOV → upsell/bundle campaign
Sales Growth %	DIVIDE([Sales]-[LY Sales],[LY Sales],0)*100	KPI with arrow	Negative growth → strategy meeting
Repeat Customer Rate	DIVIDE([Repeat Customers],[Total Customers],0)*100	Card	Below 30% → loyalty program needed

Complete KPI Formulas — Copy-Ready

```
-- ─── BASE MEASURES ──────────────────────────────────────────
Total Sales    = SUM(Orders[Sales Amount])
Total Cost     = SUM(Orders[Cost Amount])
Total Orders   = DISTINCTCOUNT(Orders[Order ID])
Total Customers = DISTINCTCOUNT(Orders[Customer ID])

-- ─── DERIVED MEASURES ───────────────────────────────────────
Total Profit   = [Total Sales] - [Total Cost]
Profit Margin % = DIVIDE([Total Profit], [Total Sales], 0) * 100
Avg Order Value = DIVIDE([Total Sales], [Total Orders], 0)

-- ─── TIME COMPARISON ────────────────────────────────────────
Last Year Sales = CALCULATE([Total Sales], SAMEPERIODLASTYEAR(Orders[Order
Date]))
Sales Growth %  = DIVIDE([Total Sales] - [Last Year Sales], [Last Year Sales],
0) * 100
YTD Sales       = TOTALYTD([Total Sales], Orders[Order Date])
```

```
-- ─── CUSTOMER MEASURES ──────────────────────────────────────
Repeat Customers = CALCULATE(
    DISTINCTCOUNT(Orders[Customer ID]),
    FILTER(
        VALUES(Orders[Customer ID]),
        CALCULATE(DISTINCTCOUNT(Orders[Order ID])) > 1
    )
)
Repeat Customer Rate = DIVIDE([Repeat Customers], [Total Customers], 0) * 100

-- ─── SHARE MEASURES ─────────────────────────────────────────
State Sales % = DIVIDE(
    [Total Sales],
    CALCULATE([Total Sales], ALL(Orders[State])),
    0
) * 100
```

SECTION 6: FILTER CONTEXT AND ROW CONTEXT

This is the most important and most misunderstood concept in all of DAX. Master this section and you will understand why DAX behaves the way it does in every situation.

The Shopkeeper Analogy

Imagine you are a shopkeeper. Someone asks: 'What is your total sales?' You say '₹5 lakh.' Then they ask: 'What about only in Maharashtra?' You say '₹2 lakh.' Same shopkeeper, same data, different answer because the question changed.

In Power BI, every visual has an invisible question — the Filter Context. Your DAX measure responds to that question automatically.

Filter Context

Filter Context is the set of active filters when DAX evaluates a formula. These filters come from: slicers, visual filters, row/column values in a matrix, and cross-filtering from other visuals.

```
Measure: Total Sales = SUM(Orders[Sales Amount])

Scenario A — No filters applied:
  Filter Context: All rows of Orders
  Result: ₹45,67,823

Scenario B — State slicer = Maharashtra:
  Filter Context: Rows where State = 'Maharashtra'
  Result: ₹12,34,500

Scenario C — Maharashtra + Category = Electronics:
  Filter Context: Maharashtra AND Electronics rows only
  Result: ₹4,56,000
```

Row Context

Row Context is active when DAX evaluates a formula row by row. This happens in Calculated Columns and Iterator functions (SUMX, AVERAGEX, FILTER).

```
-- Calculated Column: Profit
Profit = Orders[Sales Amount] - Orders[Cost Amount]

-- DAX evaluates this for each row:
-- Row 1: Sales = 2500, Cost = 1800  → Profit = 700   (stored)
-- Row 2: Sales = 5000, Cost = 3200  → Profit = 1800  (stored)
-- Row 3: Sales = 800,  Cost = 650   → Profit = 150   (stored)
-- Each row provides its own context
```

The Critical Difference

Aspect	Filter Context	Row Context
Where it occurs	Measures	Calculated Columns + Iterators

Aspect	Filter Context	Row Context
Scope	Filtered set of rows	One row at a time
Changes when	Filters change on report	Iterator moves to next row
Cross-table access	Through relationships automatically	Need RELATED() function

The Classic Beginner Mistake

⚠ COMMON MISTAKE

Writing this Calculated Column in the Orders table: Segment Name = Customers[Segment] Error! In a Calculated Column inside Orders, you are in Row Context of Orders. The Customers table is not accessible directly.

PRO TIP

Correct approach: Segment Name = RELATED(Customers[Segment]) RELATED bridges row context across related tables.

Context and % of Total — The Most Practical Example

```
-- WRONG: Always returns 100%
% of Total = DIVIDE([Total Sales], [Total Sales])
-- Both numerator and denominator are filtered to current state!

-- CORRECT: State share of national total
% of Total = DIVIDE(
    [Total Sales],
    CALCULATE([Total Sales], ALL(Orders[State])),
    0
) * 100

-- Numerator: Total Sales in current state filter context
-- Denominator: CALCULATE + ALL removes state filter → grand total
-- Result: Maharashtra = 27%, Karnataka = 18%, etc.
```

ℹ BUSINESS INSIGHT

This pattern — DIVIDE(measure, CALCULATE(measure, ALL(column))) — is the single most used DAX pattern in real dashboards. Memorize it.

Context Transition — Advanced

Context Transition happens when CALCULATE converts Row Context into Filter Context inside an Iterator. When SUMX iterates over each customer row, CALCULATE inside it converts that row identity into a filter — automatically filtering the Orders table to that customer.

```
-- Per-customer sales (SUMX iterates Customers table)
```

```
Customer Totals = SUMX(
    Customers,
    CALCULATE([Total Sales])  -- context transition happens here
)
-- For each customer row, CALCULATE filters Orders to that customer
```

SECTION 7: REAL PROJECT WALKTHROUGH

You have just joined as a Data Analyst at ShopIndia. Your manager hands you three months of sales data and says 'I need a dashboard ready by Friday.' Here is exactly how you approach this professionally.

STEP 1: Business Questions

Before touching Power BI, sit with stakeholders and ask:

7. What decisions will this dashboard support?
8. Who uses it — CEO, regional managers, product team?
9. What time periods matter most?
10. What comparisons do they need most often?

For ShopIndia, the key business questions are:

- What is total revenue, profit, and order count this quarter?
- Which states are our top performers?
- Which product categories are growing or declining?
- How does this quarter compare to last year?
- Who are our most valuable customers?
- What is our AOV trend?

STEP 2: Required KPIs

Dashboard Page	KPIs Required	Visuals
Executive Summary	Total Sales, Profit, Margin %, Orders, Customers, AOV	6 KPI Cards + 1 Line Chart
Sales Performance	Sales by State, by Category, Monthly Trend, YoY Growth	Map, Bar, Line, KPI
Customer Intelligence	Top 10 Customers, Repeat Rate, New vs Returning	Bar, Cards, Table

STEP 3: DAX Measures — Build Order

Always build in this sequence: base measures first, then derived measures, then time comparisons, then advanced measures.

```
-- ── PHASE 1: Base Measures (build these first) ──────────────────
Total Sales     = SUM(Orders[Sales Amount])
Total Cost      = SUM(Orders[Cost Amount])
Total Orders    = DISTINCTCOUNT(Orders[Order ID])
Total Customers = DISTINCTCOUNT(Orders[Customer ID])

-- ── PHASE 2: Derived from Base ──────────────────────────────────
```

```
Total Profit      = [Total Sales] - [Total Cost]
Profit Margin %   = DIVIDE([Total Profit], [Total Sales], 0) * 100
Avg Order Value   = DIVIDE([Total Sales], [Total Orders], 0)

-- ── PHASE 3: Time Intelligence ──────────────────────────────
Last Year Sales   = CALCULATE([Total Sales], SAMEPERIODLASTYEAR(Orders[Order
Date]))
Sales Growth %    = DIVIDE([Total Sales]-[Last Year Sales],[Last Year
Sales],0)*100
YTD Sales         = TOTALYTD([Total Sales], Orders[Order Date])

-- ── PHASE 4: Share and Ranking ──────────────────────────────
State Sales %     = DIVIDE([Total Sales], CALCULATE([Total Sales],
ALL(Orders[State])), 0)*100
Category Sales %  = DIVIDE([Total Sales], CALCULATE([Total Sales],
ALL(Products[Category])), 0)*100
Product Rank      = RANKX(ALL(Products[Product Name]), [Total Sales], , DESC,
Dense)
```

STEP 4: Dashboard Design Blueprint

Page 1 — Executive Overview:

- Row 1: 4 KPI cards (Total Sales | Total Profit | Total Orders | Profit Margin %)
- Row 2: Line chart (Monthly Sales + YTD line) | KPI visual (Sales Growth % vs target)
- Row 3: Bar chart (Top 5 States) | Donut chart (Sales % by Category)
- Slicers at top: Date range, State, Category

Page 2 — Regional Analysis:

- Filled Map of India: colour intensity = sales volume
- Table: State | Sales | Orders | AOV | % of Total | Rank

Page 3 — Customer Intelligence:

- Bar chart: Top 10 Customers by Sales
- Cards: Total Customers | Repeat Customer Rate
- Scatter chart: Customer count by Order Frequency

STEP 5: Insights Discovered

ℹ BUSINESS INSIGHT

Maharashtra and Karnataka together = 42% of total sales. Electronics grew 28% YoY. Fashion is flat. AOV dropped 12% Q-o-Q. Repeat Customer Rate is 23% vs 40% industry benchmark. Top 10 customers = 18% of revenue (concentration risk).

STEP 6: Recommendations

11. Launch a loyalty program targeting 2+ order customers. Target: improve repeat rate from 23% to 35%.
12. Increase inventory and campaigns in Maharashtra and Karnataka for Q4 festival season.
13. Investigate AOV decline. Consider increasing minimum order value for free delivery.
14. Protect Electronics momentum with tier-2 city targeting campaigns.
15. Flag top 10 customer accounts for dedicated relationship management to reduce concentration risk.

SECTION 8: INTERVIEW PREPARATION

These 30 questions cover everything from fresher fundamentals to scenario-based problems you will face in real Data Analyst interviews. Practice answering out loud — timing yourself at 90 seconds per answer.

Fresher Level — Questions 1 to 10

Q1: What is DAX? Where is it used?

Answer: DAX stands for Data Analysis Expressions. It is the formula language used in Power BI, Power Pivot in Excel, and SQL Server Analysis Services. I use it to create measures and calculated columns to perform calculations beyond raw data display.

Q2: What is the difference between a Measure and a Calculated Column?

Answer: A Measure is dynamic — evaluated in filter context, recalculates with every filter change, lives in a table but not physically added. A Calculated Column is static — evaluated row by row, stored in the table. Use Measures for KPI cards and charts; use Calculated Columns for row-level categorization.

Q3: What does SUM do? Give an example.

Answer: SUM aggregates all values in a numeric column. Example: Total Sales = SUM(Orders[Sales Amount]). Filtered by Maharashtra, it automatically shows only Maharashtra sales.

Q4: Why use DIVIDE instead of '/'?

Answer: DIVIDE(numerator, denominator, alternate_result) safely handles division by zero, returning the alternate result instead of an error. The / operator crashes visuals when the denominator is zero or blank.

Q5: What is DISTINCTCOUNT and when do you use it?

Answer: DISTINCTCOUNT counts unique values in a column. Use it for unique customers, unique orders, unique products — whenever you need a count of distinct entities rather than total rows.

Q6: What is the difference between FILTER and ALL?

Answer: FILTER returns a subset of rows matching a condition. ALL removes filters from a table/column, returning all rows regardless of context. FILTER adds restrictions; ALL removes them.

Q7: What is RELATED and when do you use it?

Answer: RELATED brings a single value from a parent (one-side) table into the current row context of a child (many-side) table. Used in Calculated Columns to access related table attributes, e.g., Product Category = RELATED(Products[Category]) in the Orders table.

Q8: How do you calculate Profit Margin in Power BI?

*Answer: Profit Margin % = DIVIDE(SUM(Orders[Sales Amount]) - SUM(Orders[Cost Amount]), SUM(Orders[Sales Amount]), 0) * 100*

Q9: What is SUMX and when prefer it over SUM?

*Answer: SUMX iterates over each row in a table, evaluates an expression per row, then sums results. Use SUMX when row-level computation is needed first — e.g., Total GST Revenue = SUMX(Orders, Orders[Sales Amount] * 1.18). SUM cannot compute this since the GST amount is not a stored column.*

Q10: What is a Date Table and why is it needed?

Answer: A Date Table has one row per calendar date with no gaps. It is required for time intelligence functions (TOTALYTD, SAMEPERIODLASTYEAR) to calculate period boundaries correctly. Mark it as a Date Table in Power BI and create a relationship with Orders[Order Date].

Intermediate Level — Questions 11 to 20

Q11: Explain Filter Context vs Row Context with an example.

Answer: Filter Context is active in Measures — it is the set of filters applied on the report. Row Context is active in Calculated Columns and Iterators — it evaluates one row at a time. Example: Total Sales = SUM(Orders[Sales Amount]) uses filter context. Profit = Orders[Sales Amount] - Orders[Cost Amount] as a Calculated Column uses row context.

Q12: How do you calculate percentage of total using DAX?

*Answer: Category % = DIVIDE([Total Sales], CALCULATE([Total Sales], ALL(Products[Category])), 0) * 100. ALL removes the category filter so the denominator always reflects the grand total.*

Q13: What is CALCULATE and why is it the most important DAX function?

Answer: CALCULATE evaluates an expression in a modified filter context. It can add, remove, or override filters. Almost every advanced measure uses CALCULATE. Example: Maharashtra Sales = CALCULATE([Total Sales], Orders[State] = 'Maharashtra').

Q14: How do you write a Year-over-Year growth formula?

*Answer: Last Year Sales = CALCULATE([Total Sales], SAMEPERIODLASTYEAR(Orders[Order Date])). YoY Growth % = DIVIDE([Total Sales] - [Last Year Sales], [Last Year Sales], 0) * 100.*

Q15: What is RANKX? Give a business example.

Answer: RANKX ranks a value across a table. State Sales Rank = RANKX(ALL(Orders[State]), CALCULATE(SUM(Orders[Sales Amount])), , DESC, Dense). Rank 1 = highest-selling state.

Q16: What is Context Transition?

Answer: Context Transition occurs when CALCULATE inside an Iterator converts Row Context into Filter Context. As SUMX iterates over customer rows, CALCULATE inside it automatically filters the data to that customer's rows.

Q17: Explain the difference between ALL and ALLEXCEPT.

Answer: ALL removes all filters from specified table or columns. ALLEXCEPT removes all filters EXCEPT those specified. Example: ALLEXCEPT(Orders, Orders[State]) removes all filters but keeps the State filter — useful for 'share within state' measures.

Q18: How do you find Top N customers by sales?

Answer: Use TOPN inside CALCULATE: Top 5 Sales = CALCULATE([Total Sales], TOPN(5, VALUES(Orders[Customer ID]), [Total Sales], DESC)). Combine with RANKX for ranking display.

Q19: What is USERELATIONSHIP?

Answer: USERELATIONSHIP activates an inactive relationship for a specific calculation. When Orders has both Order Date and Ship Date linked to the Date table, one is inactive. USERELATIONSHIP(Orders[Ship Date], Date[Date]) inside CALCULATE uses the ship date relationship.

Q20: How would you calculate sales of last 30 days vs previous 30 days?

Answer: Last 30 Days = CALCULATE([Total Sales], DATESINPERIOD(Orders[Order Date], TODAY(), -30, DAY)). Previous 30 Days = CALCULATE([Total Sales], DATESINPERIOD(Orders[Order Date], TODAY()-30, -30, DAY)).

Scenario-Based — Questions 21 to 30

Q21: How do you show a running/cumulative total by month?

Answer: Cumulative Sales = TOTALYTD([Total Sales], Orders[Order Date]). Display as a line on the same axis as a monthly bar chart to show trajectory toward the annual target.

Q22: What does RELATED return when there is no matching record?

Answer: RELATED returns BLANK. Handle with: IF(ISBLANK(RELATED(Products[Category])), 'Unknown', RELATED(Products[Category])).

Q23: Total Sales shows the same value in every row of your state table. What went wrong?

Answer: The measure is not responding to the row-level state filter. Likely cause: the State field in the table visual comes from a table not connected to Orders. Check relationships and ensure the State column routes through the data model correctly.

Q24: How would you build a dynamic Top N slicer?

Answer: Create a disconnected table with N values (5,10,15,20). Create measure Selected N = SELECTEDVALUE(TopNTable[N], 10). Create rank measure. Use a visual-level filter: Rank <= [Selected N]. The slicer now controls the Top N dynamically.

Q25: How do you calculate new customers each month?

Answer: Tag each customer's first order date using MIN with ALLEXCEPT. Then in the target month, DISTINCTCOUNT customers whose first order date falls within that month. This count = new customers for the month.

Q26: Category filter is applied. You want grand total in the denominator. How?

Answer: CALCULATE([Total Sales], ALL(Products[Category])) removes the category filter regardless of what is selected in the slicer. Use this as the denominator in your DIVIDE formula.

Q27: How do you calculate Average Order Value excluding returns?

Answer: Valid Orders = CALCULATE([Total Orders], Orders[Order Status] <> 'Returned'). Valid Sales = CALCULATE([Total Sales], Orders[Order Status] <> 'Returned'). AOV ex Returns = DIVIDE([Valid Sales], [Valid Orders], 0).

Q28: SAMEPERIODLASTYEAR is returning BLANK. What are the possible causes?

Answer: 1) Date table missing or not marked as Date Table. 2) No data for same period last year. 3) Relationship between Date table and Orders not set up. 4) Date filter does not include the corresponding prior-year dates in the calendar.

Q29: How do you handle a product appearing in multiple categories due to data quality issues?

Answer: First flag it in Power Query. In DAX, take the dominant category: CALCULATE(FIRSTNONBLANK(Products[Category], 1), TOPN(1, VALUES(Products[Category]), [Total Sales], DESC)).

Q30: What is the difference between a Star Schema and Snowflake Schema in Power BI?

Answer: Star Schema: Fact table (Orders) at center, flat Dimension tables around it (Products, Customers, Date). One join from fact to each dimension. Preferred in Power BI for performance. Snowflake: Dimensions are normalized into sub-dimensions (e.g., Category → Subcategory as separate table). More joins, slightly slower, more complex DAX.

SECTION 9: 30-DAY DAX LEARNING ROADMAP

Designed specifically for Indian freshers transitioning to Data Analyst roles. This roadmap assumes basic Power BI knowledge. Follow this plan and you will be interview-ready in 30 days.

WEEK 1: FOUNDATIONS (Days 1–7)

Day 1 — Setup and Dataset

📌 Task: Install Power BI Desktop. Load e-commerce dataset (3 tables). Explore relationships.

☑ Assignment: Write 5 business questions you want to answer with this data.

📖 *Revision: Section 1 of this guide.*

Day 2 — Data Model

📌 Task: Create relationships between Orders, Customers, Products. Verify in Model view.

☑ Assignment: Identify Fact table vs Dimension tables. Draw the star schema on paper.

📖 *Revision: Star schema concept.*

Day 3 — Measures vs Columns

📌 Task: Create Total Sales (Measure) and Profit column (Calculated Column).

☑ Assignment: Create Total Cost, Total Orders, Total Customers.

📖 *Revision: Re-read Section 2.*

Day 4 — Aggregations

📌 Task: Practice SUM, AVERAGE, MIN, MAX, COUNT, DISTINCTCOUNT.

☑ Assignment: Build a 6-card KPI page. Table showing Total Sales by State.

📖 *Revision: Can you explain each function from memory?*

Day 5 — Logical Functions

📌 Task: Practice IF and SWITCH.

☑ Assignment: Create Order Size column (High/Medium/Low). Create Region column with SWITCH.

📖 *Revision: Build a bar chart coloured by Region.*

Day 6 — Date Functions

📌 Task: Practice YEAR, MONTH, QUARTER, FORMAT, DATEDIFF.

☑ Assignment: Add Order Year, Month Name, Quarter columns. Build monthly sales line chart.

📖 *Revision: Verify date columns display correctly.*

Day 7 — Week 1 Review

📌 Task: Delete and recreate 3 measures from memory. Rebuild one visual from scratch.

☑ Assignment: Answer without notes: What is a Measure? Write Profit Margin % formula.

📖 *Revision: Full Section 1-3 re-read.*

WEEK 2: CORE FUNCTIONS (Days 8–14)

Day 8 — CALCULATE

📌 Task: Create 5 measures using CALCULATE. Maharashtra Sales, Electronics Sales, etc.

☑ Assignment: One CALCULATE measure per category.

📖 *Revision: Read CALCULATE theory 3 times.*

Day 9 — FILTER and ALL

📌 Task: Create filtered counts and percentage measures.

☑ Assignment: Category % of Total. State table with % of Grand Total.

📖 *Revision: Verify values add to 100%.*

Day 10 — DIVIDE and ROUND

📌 Task: Replace all / operators with DIVIDE. Add ROUND to percentage measures.

☑ Assignment: Format all % measures to 2 decimals.

📖 *Revision: Why is DIVIDE safer?*

Day 11 — RANKX

📌 Task: State Sales Rank. Product Sales Rank.

☑ Assignment: Top 5 products table. Conditional format rank column.

📖 *Revision: Dense vs Skip ranking.*

Day 12 — RELATED / RELATEDTABLE

📌 Task: Add Category and Segment columns to Orders. Add Order Count to Customers.

☑ Assignment: Segment customers by order frequency.

📖 *Revision: Verify relationships are correct.*

Day 13 — Iterators

📌 Task: SUMX for GST revenue. AVERAGEX for average profit per order.

☑ Assignment: Create 3 SUMX measures with different row-level expressions.

📖 *Revision: SUM vs SUMX distinction.*

Day 14 — Week 2 Mini Project

📌 Task: Build a complete single-page dashboard using only Weeks 1-2 knowledge.

☑ Assignment: 4 KPI cards, bar chart, line chart, table with rank, 2 slicers.

📖 *Revision: Does every measure respond correctly to slicers?*

WEEK 3: CONTEXT AND ADVANCED FUNCTIONS (Days 15–21)

Day 15 — Filter Context Deep Dive

📌 Task: Work through Section 6. Test context by watching measure values change with filters.

☑ Assignment: Write filter context explanation in your own words.

📖 *Revision: % of Total with and without ALL.*

Day 16 — Row Context Deep Dive

📌 Task: Review all Calculated Columns. Create 3 new ones with row-level logic.

☑ Assignment: Debug one incorrectly written column that should be a measure.

📖 *Revision: When does row context apply?*

Day 17 — Time Intelligence Part 1

📌 Task: Create a proper Date table using CALENDAR function. Mark it as Date Table. Build relationships.

☑ Assignment: YTD Sales. Previous Year Sales.

📖 *Revision: Why is the Date table mandatory?*

Day 18 — Time Intelligence Part 2

📌 Task: SAMEPERIODLASTYEAR, DATEADD, DATESYTD.

☑ Assignment: YoY Growth % KPI visual. Current Year vs Last Year line chart.

📖 *Revision: Does SAMEPERIODLASTYEAR return correct values?*

Day 19 — Advanced CALCULATE

📌 Task: CALCULATE with multiple filters, ALL, ALLEXCEPT.

☑ Assignment: Share within Region. % growth within category.

📖 *Revision: Context transition theory.*

Day 20 — Dynamic Measure Switching

📌 Task: Create a disconnected table. Build a slicer to toggle between Sales, Profit, Orders on one visual.

☑ Assignment: Complete the dynamic measure switcher.

📖 *Revision: This is a popular interview demonstration technique.*

Day 21 — Week 3 Revision

📌 Task: Document every measure with its purpose. Create your personal DAX cheat sheet.

☑ Assignment: Review all 30 interview questions.

📖 *Revision: Full Section 6 re-read.*

WEEK 4: PROJECTS AND INTERVIEW PREPARATION (Days 22–30)

Day 22 — Full Project Setup

📌 Task: Begin ShopIndia project from Section 7. Load data. Set up relationships. Date table.

☑ Assignment: All base measures created and verified.

📖 *Revision: Section 7 Step 1 and 2.*

Day 23 — Build All 10 KPIs

📌 Task: Create all measures from Section 5. Test each measure.

☑ Assignment: Every KPI shows correct value. SAMEPERIODLASTYEAR works.

📖 *Revision: Section 5 formulas.*

Day 24 — Executive Summary Page

📌 Task: Build Page 1: 4 KPI cards, monthly trend line, growth KPI visual, slicers.

☑ Assignment: All visuals update correctly when slicers change.

📖 *Revision: Visual design principles.*

Day 25 — Regional Analysis Page

📌 Task: Build Page 2: Filled India map, state performance table with rank.

☑ Assignment: Map shows colour variation by sales. Table sortable.

📖 *Revision: Map visual configuration.*

Day 26 — Customer Intelligence Page

📌 Task: Build Page 3: Top 10 customers bar chart, repeat customer rate card.

☑ Assignment: Customer visuals filter correctly from executive page.

📖 *Revision: Cross-page filtering.*

Day 27 — Dashboard QA and Insights

📌 Task: Format all visuals consistently. Verify all measures. Document 5 business insights.

☑ Assignment: Written insight document ready.

📖 *Revision: Section 7 Step 5.*

Day 28 — Interview Prep Part 1

📌 Task: Answer Q1–Q15 from Section 8 without notes. Time each at 90 seconds.

☑ Assignment: Record answers. Identify weak areas.

📖 *Revision: Strengthen gaps from today.*

Day 29 — Interview Prep Part 2

📌 Task: Answer Q16–Q30. Practice project presentation as if in an interview.

☑ Assignment: Can you explain your entire dashboard in 5 minutes confidently?

📖 *Revision: Review all context concepts.*

Day 30 — Portfolio Finalization

📌 Task: Polish project. Export screenshots. Write one-page project summary document.

☑ Assignment: Portfolio piece ready: Business Problem, Approach, DAX Measures Used, Insights.

📖 *Revision: Final self-review of all 9 sections.*

After Day 30 — Keep Growing

- Kaggle: Find Indian retail/sales datasets. Build dashboards on new data.
- YouTube: Follow Chandoo, Guy in a Cube, SQLBI for advanced techniques.
- Power BI Community forums: Answer others' questions to deepen understanding.
- GitHub: Share your .pbix files to build credibility.
- LinkedIn: Post your dashboard screenshots with a short insight caption — recruiters notice.

> **ℹ BUSINESS INSIGHT**
> A Data Analyst who can walk into an interview with a live Power BI dashboard demonstrating real business insights is in the top 5% of candidates. Build it. Own it. Explain it confidently.

End of Guide

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